

SSHVisual: Visual Tool to Analyze SSH Data

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Abstract

[CONTEXT/BACKGROUND] Network operators use visualizations to detect attacks on their systems, exploring and interacting with data to uncover patterns that would otherwise stay hidden. Operators then use these findings to block attackers and to expand the knowledge base of known threats. Detecting anomalies and uncovering deeper patterns iteratively in data is at best, tedious and at worst, not achievable without data visualizations.

[EXISTING PROBLEM] Previous works in network visualization analyzed the IP and AS dimensions of SSH logins without much focus on the username aspect of the attackers' attempts, detected anomalous network events only where baseline events were available, and enabled operators to share findings with one another. Little attention has been paid to whether attackers observed at different times represent the same persistent actor.

[OUR SOLUTION] We present an iterative visualization SSHVisual, using login attempts collected on our own SSH-facing systems. SSHVisual analyzes the IP, AS, and username dimensions of SSH attacks, in service of SSH campaign detection. It complements automated detection by allowing operators to find anomalous events without baseline events. It correlates SSH attacks with real-world events, and it introduces a similarity metric that quantifies attacker persistence over time. Like previous systems, SSHVisual supports iterative and collaborative sharing of findings between operators.

[EVALUATION AND CONCLUSION] We evaluated SSHVisual in a qualitative study with network operators and found that it helps them block attacks and uncover patterns. Participants identified campaigns in which a set of attempted usernames recurred across unrelated IPs and ASes, and reported that this activity would not have surfaced in their existing tooling. The similarity metric surfaced attackers returning across different time periods despite changing network origin. SSHVisual shows that treating usernames as a first-class analytic dimension reveals attacker coordination that IP-centric analysis misses, and that exploratory, operator-driven detection remains valuable precisely where automated methods are weakest. [THIS PARAGRAPH CAN CHANGE IF WE DECIDE TO EVALUATE IT OURSELVES]

I Introduction

Computer systems worldwide employ SSH [1] to allow their users to securely connect to them in order to execute com-

mands and do remote administration. A lot of users do a username, password combination (password based authentication) in order to gain entry to the SSH system. Attackers, who are adversarial in nature, can use brute force attack mechanisms [2] to get into SSH systems. Sometimes, they guess what the username, password combinations are and other times, they take advantage of poorly secured systems to build botnets [2] to carry out further attacks [3, 4] or steal resources from the systems, like cryptomining [5, 6].

Given the prevalence of SSH attacks in the IPv4 space [2], we collect SSH login data on our own systems and present SSHVisual. SSHVisual serves two kinds of network operators. Network administrators, who handle day-to-day SSH traffic, need to identify attackers quickly and block them. Security analysts, who study traffic over longer time periods, look for patterns that are not immediately apparent and share what they find across organizations. Consider a single time period: an administrator can isolate the highest-volume attacker and block it immediately, while an analyst examining the same window can iteratively surface a set of low-frequency attackers that attempt an identical sequence of usernames, and group them as a coordinated botnet.

Surveys of network visualization [7, 8, 9, 10, 11] establish the maturity of the field and the gaps we target. Our contributions follow from treating the username as an attacker identity. First, actor-level analysis: we treat the attacker as a first-class actor — including the username identity, not only the source IP or autonomous system (AS) — and we identify coordinated campaigns. Since our data affords no stable baseline for automated detection, this identification is necessarily exploratory (therefore, not antagonistic but complementary to automated detection with stable baselines) — operators recognize coordination visually rather than specifying it in advance. Second, a similarity metric that quantifies attacker similarity and lets operators track how actors evolve in the username dimension across time periods. Third, correlation of actor behavior with real-world events, which is more tractable through usernames than IPs, given the limitations of IP corroboration. Additionally, following established practices, we allow for iterative operator driven collaboration and labeling of irregularities found in the data.

The rest of the paper is as follows: Section II is Related Work, Section III is Task Requirements (WOULD IT BE FAIR THAT WE ESTABLISH THE REQUIREMENTS WITHOUT CONSULTING ANY NETWORK OPERATORS?), Section IV is Design, Section V is Evaluation and Section VI is Conclusion and Future Work.

II Related Work

A long line of tools [12, 13, 14, 15, 16, 17, 18, 19] render network traffic and security events navigable for operators and analysts with the majority being network flow and traffic oriented. Additionally, a smaller alert- and log-oriented group [20, 21, 22, 23, 24, 25, 26, 27, 28, 29] organizes IDS output and heterogeneous logs of which [22] contributes a visual analysis (of the CICIDS2017 dataset) that isolates attack clusters — including SSH brute-force (SSH-Patator) — through clustering and dimensionality reduction, namely a multi-resolution wavelet-feature analysis.

A Attack Campaigns with Time

Honeypot literature does not promote the username identity as an actor in its own right (for attack campaign detection purposes): it remains an attribute to count, never a unit to track — the unit that, alongside the IP and AS, anchors our analysis. Some papers [30, 31, 32, 33, 34, 35, 36] collect and chart honeypot traffic, with several of them capturing SSH credentials (usernames and passwords). Where credentials are charted — the SCADA SSH honeypot [33], kippo-graph [34], and the container deployment [35] — usernames and passwords appear only as static top-N tallies (kippo-graph’s top-10 username and password bars are the archetype), never placed on a time axis even where the same tool charts attack volume over time. The WAN honeypot framework [37] is the closest prior system on the credential axis: it computes an IP-to-IP dictionary-overlap (containment) score to probe whether six same-/24 source IPs share a wordlist — a manual, one-off prototype of credential-based coordination analysis, performed on the IP unit and without a temporal dimension.

B Attacker Similarity with Time

Computed similarity between entities is well established in literature [38, 39, 40, 41]. We do not claim similarity computation itself as novel; the contribution lies in the entity it operates on and how it moves through time. On the entity: these systems cluster internal hosts [38], behavioral profiles [40], IP/port members [41], or internal enterprise users grouped by shared resources — not external attacker identities [39]. None treats the username as an attacker actor. On time: [40, 41] reach into the temporal dimension, but both do not compute the similarity of the attacker population across consecutive windows. We apply similarity based metrics — for example, Jaccard overlap of attackers between successive windows — particularly to the username actor unit, capturing how the attacker population persists or churns over time.

C Limits of automated detection

One premise of our approach is that SSH traffic does not admit the “normal baseline” that automated anomaly and campaign detection presuppose. The broader literature documents why such detection is fragile: Sommer and Paxson [42] note that,

despite extensive research, machine-learning anomaly detection is rarely deployed for intrusion detection, given the enormous variability of “normal,” the high cost of errors, and the difficulty of sound evaluation; Axelsson [43] shows that automated anomaly detection is throttled by the base-rate fallacy: where intrusions are rare relative to benign traffic, the false-positive rate becomes the limiting factor. Honeypot traffic departs from that setting entirely — it offers no stable “normal” against which to define an anomaly, with Gates and Taylor [44] supporting our approach by challenging the paradigm directly, questioning that anomalies and attacks are rare, that a stable normal exists and sometimes normal traffic may be malicious and malicious traffic may be normal, finally observing that detection ultimately devolves to an operator manually investigating clusters of activity.

This fragility is visible from some of the papers referenced so far. The automated detectors there are built on a per-actor baseline: Situ [38] learns a probabilistic model of each IP’s behavior and flags low-probability events, and CLIQUE [45] scores each actor’s deviation from a calculated baseline of its own expected activity. Both presume the stable per-actor “normal” that honeypot traffic — near-uniformly hostile and constantly churning — cannot supply, so they do not transfer to this setting by construction. And even where such a baseline does hold, Situ itself concedes that automation is not enough: having reviewed the limits of signature-, supervised-, and reputation-based detection, it argues that operators “do not want another automated tool with algorithms they do not trust,” and builds a human-in-the-loop visual analytics system instead. We reach the same conclusion from a starker starting point: lacking even the per-actor baseline these systems assume, we treat anomaly and campaign detection as operator-driven from the outset, providing visual support for it rather than another automated detector.

D Actor identity and real-world event correlation

The attacks we observe do not arrive at a steady rate; they rise, fall, and change in character as events unfold in the wider world, so understanding them means lining them up against what is happening outside the network. The link is well documented: Bilge and Dumitras [46] found that the number of attacks against a given vulnerability can jump by up to five orders of magnitude — a hundred-thousand-fold increase — once it is publicly disclosed, and Sabottke et al. [47] showed that future exploitation can be predicted from signals in disclosure channels alone. Allodi and Massacci [48] add that a vulnerability’s official severity score is a poor guide to whether it is actually attacked, so the dependable way to learn what attackers are reacting to is to watch real activity and line it up against real events, not to read it off a score. We want to do this at the level of a campaign — a group of attackers acting together — rather than one attacker at a time, and that raises a practical problem: a campaign rarely shows up as a single source address. IP addresses do not stay tied to one machine — one host is assigned many addresses over time [49], hosts

cycle through new addresses [50], and counting by address can overstate the number of real machines by orders of magnitude [51] — so it is likely that a view organized by address scatters one campaign across dozens of entries that look unrelated. The username pulls those pieces back together: the same login names likely recur across the changing addresses, so grouping by username re-assembles the campaign that an address-by-address view splits apart. A username is not a perfect identity either — names like "root" are tried by almost everyone, and one campaign runs through a whole list of names — so we treat it as one analysis dimension alongside the address and the network it belongs to (its AS), resolving the overlaps with the similarity and grouping methods above.

E Operator labeling and collaboration

Since detection on our data is operator-driven, the findings an operator produces by hand must be captured and shared rather than discarded. In two lines of work [52, 18], interactive labeling has been used to fold the operator’s judgments back into the system. Collaborative analysis has been supported through shared artifacts with [16] and [19] being notable examples. These systems establish that labeling and collaboration are valuable, which is why our labeling-and-sharing mechanism is the step that turns an identified set of usernames into a durable, shared campaign.

III Task Requirements

A Data and Domain Characterization

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B Task Abstraction

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Table 1: Example placeholder table

Requirement	Task 1	Task 2	Task 3	Task 4
Row A	+	-	+	+
Row B	tech	plan	-	+
Row C	+	+	+	-



Figure 1: Show campaign detection using username+IP+AS

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IV Approach

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A First Component

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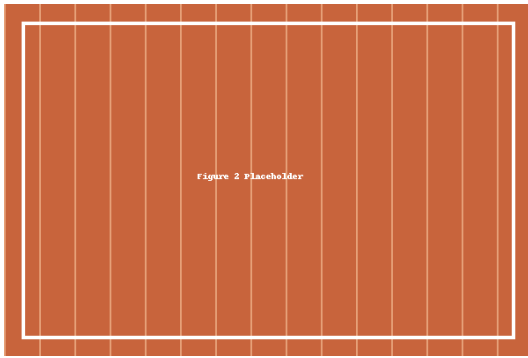


Figure 2: Show collaboration + labeling

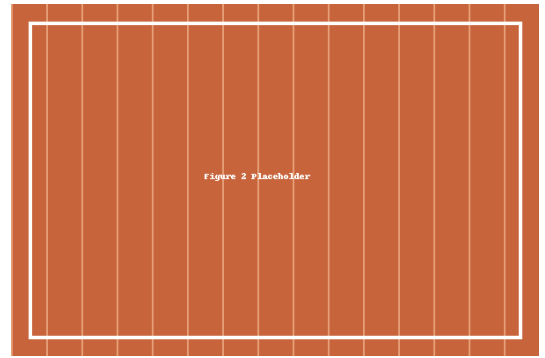


Figure 3: Show similarity metric at play

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B Second Component

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C Third Component

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V Implementation

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A System Architecture

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B Data Pipeline

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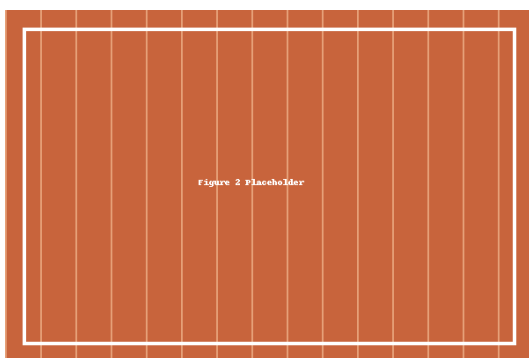


Figure 4: Show correlation of attacks with real events

VI Evaluation and Discussion

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A Case Studies

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B Discussion

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VII Summary

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B Future Work

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Acknowledgments

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